

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented RTL for high-speed **Ethernet MAC** and **PCS IP cores** up to 400Gb, adhering to IEEE 802.3dj specifications.
- Collaborated with verification and back-end engineers daily within a co-located team to ensure design integrity and timely delivery of IP blocks.
- Applied digital design best practices, including **synthesis optimization** and **timing closure** analysis for multi-rate, multi-port architectures.
- Developed Python scripts to automate data extraction and reporting for internal system monitoring, reducing manual effort by approximately **15%**.
- Managed production support for enterprise Java/.NET applications, responding to P1/P2 incidents and maintaining service level agreements using **Splunk** and **Dynatrace**.
- Contributed to the development of REST APIs using **Java Spring Boot** for cross-service data exchange, implementing robust error handling and validation.
- Containerized backend services using **Docker** and deployed them to an **AWS EC2** environment managed via Jenkins CI/CD pipelines.

SKILLS

Hardware Description Languages – Verilog, SystemVerilog

Digital Design – RTL Design, State Machines, Clocking, Timing Closure, Synthesis

Networking Standards – Ethernet, IEEE 802.3dj, High-Speed Networking

Software Development – Java, Spring Boot, Python, REST APIs, SQL

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Linux Shell Scripting

PROJECTS

High-Speed Ethernet IP Design and Integration

- Developed RTL for **Ethernet MAC** and **PCS IP cores** targeting speeds up to 800Gb, focusing on modular design for multi-port configurations.
- Collaborated with verification engineers to integrate testbenches and debug RTL issues, ensuring compliance with **IEEE 802.3dj** standards.
- Participated in design reviews and contributed to the refinement of synthesis constraints to achieve target timing closure metrics for critical paths.

Automated Data Pipeline for System Health Monitoring

- Engineered Python scripts to extract and process log data from production servers, creating automated reports on system health and performance.
- Developed SQL queries against PostgreSQL to validate data integrity and aggregate metrics for trend analysis, supporting incident root cause identification.
- Integrated the reporting script into a scheduled Jenkins job, providing weekly insights on application stability to operations teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)